

Deep Learning Based Sign Language Gesture Recognition with Multilingual Translation

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Abstract— The new developments and recent research in the field of machine translation and image captioning have enhanced the job of translation using machines. Sign language which has been a barrier in communication has successfully included itself as a major language. Sign language translation to and from becomes of utmost importance and image captioning helps the cause. The project includes various models including YOLOv5, LSTM-GRU and random forest with libraries like OpenCV and frameworks like MediaPipe to attain the required goal. Machine translation is done in multiple languages namely English, Hindi and French. Machine Translation is attained through a combination of different tokenizers, optimizers and models like auto-tokenizer, Adam optimizer and LSTM respectively. This project would not only help bridge the gap of communication eradicating the language barriers but also help in efficient communication with visually impaired people using text to speech and also with the auditory impaired individuals by text-to-speech and also using image recognition to understand sign language eradicating almost all the barriers that are present in verbal communication with the impaired. To create such a model, an individual needs to understand the scope of the system and also the requirements and a proper understanding of the entire procedure which accumulates into building an efficient system that suits every basic requirement of the user and also the goals the system is set to achieve.

Keywords— American Sign Language, Random Forest, Long Short-Term Memory, Recurrent Neural Network, Grated Recurrent Unit, Automated Audio Captioning, Automated Speech Recognition, Natural Language Processing, Natural Language Tool Kit, Bi-Directional Bi Lingual Evaluation Understudy.

I. INTRODUCTION

Voice is the most fundamental, widespread, and efficient way for individuals to engage with one another. Speech technologies are widely accessible for a restricted but engaging set of tasks. Machines are able to respond appropriately and reliably to human voices, and also deliver valued and useful services. Such systems are preferred by humans since speaking is faster than typing or any manual input. Because spoken language dominates human communication, it is reasonable for people to anticipate speech interactions with computers. A speech-to-text system can be established, which allows a computer to interpret voice requests and dictation into text.

Deaf and dumb individuals have their own sign language, which communicates through facial expressions, body movements, and hand gestures. Nonverbal communication is utilised for social engagement and communication with

family members. Sign languages differ by nation and area, and each has its unique vocabulary and grammar. Hand configuration recognition, motion discrimination, face expression recognition, and linguistically meaningful spatial contrasts are all required. Interaction is critical in sign language software applications because it converts users from passive recipients to active participants in the learning process.

II. PROBLEM STATEMENT

The Deaf and Hard of Hearing (DHH) community have been lagging the communication race since a long time and to give them a head start and also break the communication barrier between various rigid cultures throughout the world, a common base system is required. The problems are not only inclined towards the acceptance of the system but also the efficiency of the system which based on the various evaluation metrics should perform well on human biases as well.

III. LITERATURE REVIEW

Neural networks have significantly advanced machine translation, enabling automatic translation of diverse languages and bridging communication gaps worldwide. Machine translation now includes speech-to-text conversion and live captioning. Live captioning, similar to subtitles, is saved for simultaneous translation to speech using neural networks.

Deaf and mute individuals use sign language, which relies on facial expressions, body movements, and hand gestures for communication. Sign language, can be integrated into apps with video to aid hearing-impaired users. There are two types: Isolated Sign Language (single gestures) and Continuous Sign Language (phrases). Computer vision and image processing techniques enable gesture recognition from videos or images, enhancing human-computer interaction.

A. Sign Language Recognition

This comprehensive research explores various models for sign language recognition, leveraging deep neural networks and computer vision techniques to enhance communication for individuals with speech and hearing impairments. One study presents a two-way conversation system using deep neural networks and edge detection, achieving 90% accuracy with MediaPipe and LSTM-based neural networks as in [1]. Another model utilizes YOLOv5 for sign gesture recognition, reaching an 88.4% accuracy, outperforming CNN in real-

time detection as in [2]. A third approach combines LSTM and GRU for Indian Sign Language, achieving 97% accuracy mentioned in [3]. Additionally, a model using MediaPipe Framework and KNN predicts ASL alphabet characters with 86-91% accuracy ([6]). Lastly, a hand gesture recognition system employing LSTM and MediaPipe landmarks achieves 90% accuracy, demonstrating the potential for real-time application as studied in [7]. These studies highlight advancements in gesture recognition, emphasizing the effectiveness of neural networks and computer vision in creating accessible communication tools.

The paper [28] examines the potential of large language models (LLMs) for sign language translation (SLT), focusing on the NIASL2021 dataset of Korean and Korean Sign Language (KSL) sentence pairs. It introduces Ko-GPT-Trinity-1.2B+VS, a GPT-3-based SLT model with vocabulary sharing, which achieves impressive BLEU-4 scores of 22.06 in text-to-gloss (T2G) and 45.89 in gloss-to-text (G2T) tasks. The research highlights the advantages of LLMs with vocabulary sharing in mitigating the resource scarcity problem in SLT.

B. Audio Captioning

The study [10] describes a novel technique to automatic audio captioning that employs Transfer Learning and Reconstruction Latent Space Similarity Regularisation (RLSSR), a unique self-supervised goal that augments model training by maximising the similarity between encoder and decoder embedding. Automated Audio Captioning is a new study topic that tries to characterise events in an audio clip's foreground and background. Convolutional neural networks are a well-liked option for encoding information from audio input. However, they are not sufficient for Automated Audio Captioning because the model needs to capture both overarching events (like a car running in the background) and local events (like a loud honk) that happen in shorter timeframes. They suggest using a transformer encoder, which consists of many layers of feedforward and multi-head self-attention, in addition to the convolutional encoder to get around this. The authors propose a method for improving model performance by combining convolutional and transformer encoders, transfer learning, and a self-supervised learning objective called Reconstruction Latent Space Similarity Regularisation (RLSSR). RLSSR reduces the Euclidean distance between the decoder layer's reconstruction in latent space and the encoder's output. The authors use the transformer encoder in their encoder and show that using this objective can outperform state-of-the-art results on the Clotho dataset. Their method was tested on the Clotho audio captioning dataset, achieving a word error rate (WER) of 23.9%, a significant improvement from the state-of-the-art 28.6%. The authors' method might pave the way for new directions in automatic audio captioning research. It shows how the learning objectives of self-supervised learning and PANN transfer learning can be used to improve the performance of the audio captioning model.

The study [11] addresses the challenge of Automated Audio Captioning (AAC), which generates natural language descriptions for audio inputs. Due to limited and costly audio-caption datasets, AAC model performance is often subpar. The researchers propose FeatureCut, an online data

augmentation method that enhances training data by combining encoder-decoder structures. FeatureCut utilizes attention-weight maps from the language decoder to create a Feature Attention-weight Table (FAT), identifying important acoustic features. These features are progressively removed to improve model learning, and Kullback-Leibler divergence ensures balanced learning capacities. Tested on the Clotho dataset, FeatureCut boosts CIDEr scores by 3.1% and 5.1% in CNN-Transformer and MAAC models, respectively, offering a significant performance improvement in AAC.

C. Machine Translation

An interactive system utilizing Automatic Speech Recognition and Machine Translation for learning Hindi as a Second Language was developed by the authors in [13] to overcome the challenge of converting English sentences to Hindi. Because the language may be learnt more quickly through repeated exposure to its vocabulary and phrases, the approach has outperformed traditional methods in terms of learning speed. The system's overall accuracy of around 67% can initially benefit learners of second languages.

This paper [14] proposes a Machine Translation technique based on Neural approach- based quality estimation technique for improving translation of English to Hindi. The technique uses DL-based quality estimation feature extraction and experiments on multiple features along with various Neural RNN models. The recurrent layer in the neural network generates a report on how well each part of the sentence fits in the given sentence and the efficiency of the translator. The accuracy and loss of all predicted RNN models namely, LSTM 256, LSTM 128, Bi-LSTM, Bi-GRU 256, GRU 128, etc are compared. Also, the evaluation matrices of the same are compared and the result are displayed in tabular form in the given paper. The paper applies all of these models on the English to Hindi Translation Dataset and used the same WSJ dataset to compare the results of all the models. This technique differs from translation evaluation because it is not dependent on the reference translation. Similar to how humans read and understand words in a sentence by processing the contextual information available, the proposed model uses RNN encoders to convert the given input sentence into internal representations. It then bases its final prediction decision on the contextual information in the sentence.

This paper [15] presents a framework for German-English machine translation using GRU RNN as a deep learning GRU-based Recurrent Neural Network (RNN) modelled as a framework. The English-German dataset from WMT2021, which includes 400,000 strings from German news with parallel English translations, serves as the foundation for the system. The provided framework lays the groundwork for translating strings from German to English and creates avenues for more research in the field. Efficiency always serves like the ultimate objective of any system and the authors have achieved the maxima in this approach. EMNLP constitutes of a dataset of sentences from news articles in pairs of multiple languages as a parallel translated corpus. The work pushes the need of further research in the given topic by giving a brief on the latest performance scores as a comparison matrix. It also addresses the various challenges present in MT of various sentence generation systems. The authors experiment the German- English news translation by

modelling a GRU RNN deep learning framework on a parallel German-English translated news corpus. The same framework can be applied to language pairs with parallel corpus and can be used to generate higher rates of efficiency and human-intuition-level sentences.

In some cases, there fails to exist a corpus that is parallel to 2 languages, in such cases the idea of a bridge language can do great favours to a project since at times there is a language which can be connected to two languages and the corpus is also easily available. Especially for Statistical Machine Translation (SMT) the corpus plays a very important role and to not have a strong corpus for a bilingual set of languages as researched by the authors in [16]. This idea is useful in situations where there are sufficient resources between pairs of languages, such as German and Farsi. The bridge language idea has been tested before for improving word alignment and enhancing results but it also would work equally for cross-language retrieval and translation, pushing through the advantage of running new SMT systems for language sets with no parallel corpus. The first method is to bridge between pair languages Farsi and German at translation time, using the

as the pivot language. The second method uses the direct Farsi-German corpus, which has shown promising results in combining these two machine translation systems.

IV. METHODOLOGY AND IMPLEMENTATION

A. Proposed System Model

In the block diagram shown in Fig 1, as we can see that the input can be taken from an attached video recording system(camera) or a microphone to record the audio, these inputs are then processed and converted to captions or text in the bridge language (English), these texts are then processed to either speech output in English or displayed as captions according to the user requirement. The option also available to the users is that they could either convert the output from English to other widely spoken languages namely Spanish, French, German or Hindi. The system not only gives the output in the form of these languages but also accepts inputs in these respective languages.

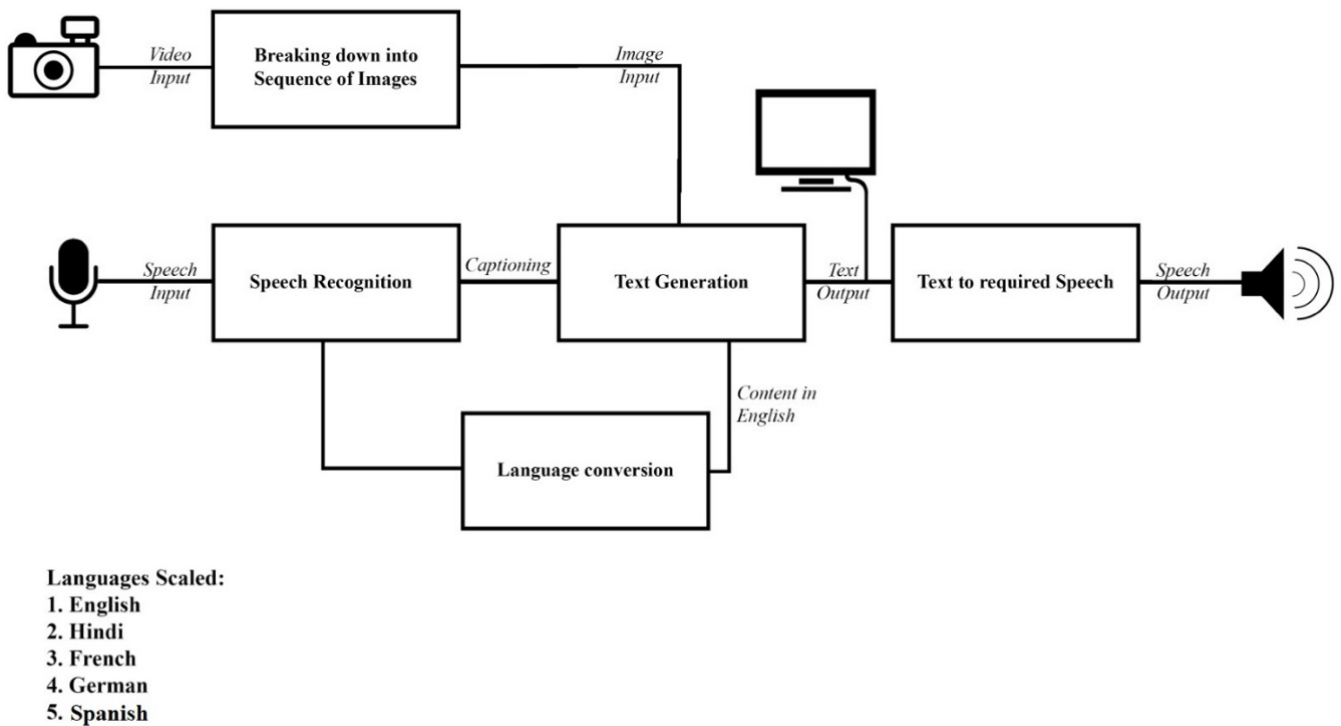


Fig 1 Proposed System Model

English language as the pivot language. The bridge language idea, which combines Farsi and German at translation time, has been applied to other languages like English, French, and German. This method has shown promising results in improving word alignment, cross-language retrieval, and running new SMT systems for languages without parallel corpus. The first method bridges between Farsi and German at translation time, using English

The sign language detection module captures the user's gestures in real-time breaking down the video into a sequence of images and provides the corresponding text output for the input sign language gesture. The SSD MobileNetV2 Classification model [3], [6] or the YOLOv5 model [4], can be used here which surpasses CNN in accurate real-time gesture detection. The InceptionResNetV2 combined with LSTM-GRU model [5] can also be used which gives good

results on real-time detection. The text output generated can be then converted to required language using the machine translation module and in speech format as well.

ASR tasks can generate efficient speech transcripts and AAC helps with descriptive audio noises, but keeping in mind that we have sign language as one of our languages it becomes very important to take in to consideration the descriptive noise and keep the caption as precise as possible. Languages like mandarin generate words which at times get accounted as noises or extended notes while speaking which could affect the translation on the opposite end. Therefore, we propose a joint modelling of ASR and AAC tasks [8] which can not only model the required data but also give the efficiency output of both ASR and AAC tasks. These joined together give rise to an extensive experiment over noisy speech samples with different levels of background noise and proves that it cannot only get the better hold of independently trained ASR and AAC tasks.

As we move on our focus to translation from detection a proposed model LSTM with 256 layers produced best results for translation [10]. LSTM with 256 layers not only surpasses all other models but also gives close to human level precision values in most cases. The datasets that we propose to use are, ASL-American Sign Language [2], [6], IISL2020-Indian Sign Language [5], WSJ with 37K Speech Samples [8], AudioCaps 26k Non-Speech Samples [8], News Commentary dataset obtained on the EMNLP 2021[11], and Human-annotated Chinese polyphonic character dataset with 859,112 unique sentences [13]. These datasets include enough training set for Sign Language Detections, Audio Captioning and Translation to and from Hindi, German and Mandarin to English Language.

V. RESULTS

A. Sign Language Detection

We performed sign language recognition on the ASL Alphabet Dataset using Random Forest Classifier, details of which are mentioned in the Methodology section. We obtain approximately 97% accuracy on the test data and achieve nearly accurate results on live feed detection as well. The real time results captured are as shown in Fig 3.

B. Machine Translation

We implemented the custom architecture of our model on the given set of languages and the results were generated and compared with all the other languages and different types of approaches that we could possibly try in machine translation. These outputs can be seen in the interface as in Fig 3.

C. Overall summary for different functionalities with various approaches

Table 1 shows the performance analysis and summary of implementation of different functionalities like sign language detection, audio captioning and machine translation using various approaches. We see that sign language detection gives best results when Random Forest algorithm is used. For Audio captioning, built-in speech recognition module works the best. In case of machine translation different approaches for various translations are also summarised.



Fig 3 Real time results of sign language detection

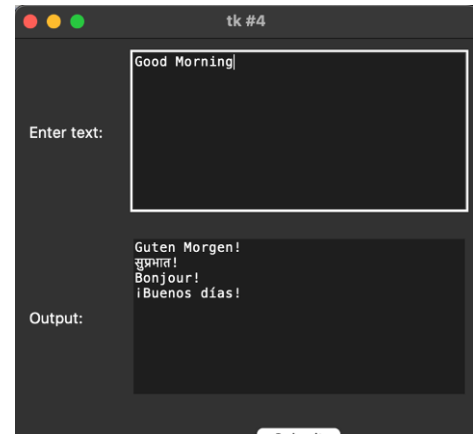


Fig 2 Machine translation interface with conversion of English in four different languages

Table 1 Performance analysis of various algorithms

Feature	Algorithm/Hyperparameters altered	Results (Accuracy)	Inference
Sign Language Detection	- YOLO - LSTM-GRU - Random Forest	88.4% 97% 98%	- No continuous recognition. - Computationally Heavy - Works poorly on dynamic datasets.
Audio Captioning	- AAC - Double Decoder - Inbuilt Speech Recognition Module.	80%* 92%* 98%*	- Background Noises are also captioned as words - Background noises are ignored but computationally heavy - Easy to install and interpret
English-French	- Pre-Built Models (LLM) T5 - Deep Learning Models (Bi-LSTM encoder and LSTM decoder)	97% 96.94%	- LLMs are trained for specific tasks and re-training such models with custom parameters require a lot of computation power. - The various combination of layers give results with the best possible result for the given pair of encoder decoder considers context as well as computationally efficient.
English - Hindi	- Word by Word Dataset - IITB Parallel Corpus	73% 96%	- Fails to capture the context of the words - A dataset of most commonly used jargons but fails to capture any latest language slangs.
English-Spanish	- AutoTokenizer - Bag-Of-Words	96.03% 75.03	- It takes time to understand the tokenizing sequence but gives the best output. - Most of the times fails to understand the semantic reference.
English-German	Deep – Learning Model – Parallel Corpus – Cross Attention – AutoTokenizer – Adam Optimizer	78%*	This shows that the combination of models that worked the best for other languages might not work as great for German Language therefore it is of utmost importance to understand the task at hand before deciding on the model architecture and peripherals.

* Indicates BLEU score

VII. SCOPE AND FUTURE WORK

The paper delves into Machine Translation's (MT) diverse applications, spanning education, communication, public services, customer support, and media/entertainment. MT enables real-time access to educational resources for the differently-abled, facilitates inclusive communication between individuals, and improves accessibility in online meetings. It enhances public services by enabling effective communication in deaf and dumb schools and offers accessible customer support through language translation and speech modules. MT's Audio Captioning Module provides subtitles for media content, breaking language barriers. The paper highlights future prospects with Language Learning Models (LLMs) like Llama, indicating ongoing advancements in the field.

The future scope of this research is to build upon the accuracies that we have displayed and get better accuracies with different methodologies and additions. The attention layer for example could improve the results to a good extent and therefore, further advances that could hold the context and understand the semantical aspect of the text could perform much better and the results could be improved. The files are comparatively heavier to execute and work much better as stand-alone programs than compiled together. These are some aspects which can attract future work and prove as more efficient and accurate results.

VIII. CONCLUSION

In this project, we have implemented various classification algorithms and deep learning models which utilised for sign language recognition. Our observation is that Random Forest emerged as one of the most computationally efficient method for Sign Language Detection and interpretation. The second mode of input that we have for our system is speech. The system should be robust and immune to disturbances in the background and multiple users speaking at the same time. Speech recognition module from python libraries perform this task with 94% accuracy which cannot be beaten by a transformer trained at the Clotho dataset or the AudioCaps dataset. Finally the Machine translation segment provides a wide range of results comparative to the various models that we evaluate and deciding upon the best accuracy ranges and the easier accessibility of the model we choose upon the best model specific to language requirements and compile in the application. The system uses an LSTM based model which outperforms google translate on human observed precision and a BLEU score or 96%. The system is attempted on Android studio using the Machine learning models extension of android studio. The system is also accessible based on a tkinter platform creating a WebApp. Therefore, making it accessible to the vast variety of users looking to not only develop but also use machine translation.

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